

# Ethan Tran

🌐 Portfolio | 🗣️ stutterk1d | 🌐 Ethan Tran | ✉️ ethantran03@gmail.com | 📞 +714-234-8860

## Education

### University of California - Irvine

Bachelor of Science in Data Science

Irvine, California

September 2024 – June 2026

### University of California - Riverside

Bachelor of Science in Data Science

Riverside, California

September 2022 – June 2024

**Relevant Coursework:** Data Structures & Algorithms, Discrete Structures, Data Analysis Methods, Software Construction, Linear Algebra, Database Management, Statistics & Probability, Statistical Computing with R

## Technical Skills

**Languages:** JavaScript, TypeScript, React, Python, C++, HTML, CSS, SQL, MATLAB, R

**Libraries:** TensorFlow, SciPy, Matplotlib, Pandas, NumPy

**Tools:** Git, Github, PyTest, GTest, PowerBi, Google Cloud Platform, Tableau, Jupyter, Excel

## Experience

### ITTIA

Data Science Intern

Irvine, CA

May 2025 – September 2025

- Build **on-device** data pipelines with ITTIA DB Lite on STM32 to collect, filter, and aggregate time-series sensor data for **feature engineering** and **machine learning**.
- Designed **schemas**, wrote **queries**, and benchmarked ingestion/latency to support low-memory, low-latency **analytics**.
- Authored examples for embedded **ML workflows** that demonstrated reproducible data prep and evaluation on microcontrollers.

### Crawlbee

Research Fellow

Remote

June 2023 – August 2023

- Built a custom **data pipeline** and **MySQL** backend to process over **100 million monthly web crawls**, beating performance targets by 70%.
- Developed a **real-time data visualization dashboard** using **Grafana** and **AWS EC2** to help founders track metrics and allocate resources.
- Automated deployment and testing with **Docker** and **GitHub Actions**, maintaining **133 ms average latency** under high traffic.

### The Ballard Cognitive and Neural Sciences Lab

Research Assistant

Riverside, CA

September 2023 – June 2024

- Analyzed fMRI data** to investigate the neural basis of decision-making, contributing to cutting-edge neuroscience research.
- Conducted **data preprocessing, feature engineering, and ROI selection** using FSL and Python.
- Developed statistical models and applied contrasts and t-contrasts, resulting in the identification of key activation patterns.
- Visualized findings through statistical parametric maps and ROI plots, creating insights for the research team.

### Data Science Academy

Academic Leader

Riverside, CA

March 2023 – June 2024

- Instructed over **200 middle and high school students** in data science fundamentals, fostering early interest in STEM fields.
- Designed and adapted curriculum covering Python programming, data cleaning, visualization, and machine learning using Google Colab.
- Implemented interactive teaching methods, resulting in a 30% increase in student engagement and positive feedback shown through surveys.
- Mentored students in developing mini-projects, building their practical understanding of data science concepts.

### SMART Program

Mentee

Riverside, CA

January 2024 – June 2024

- Analyzed a **918-patient heart disease dataset** with a graduate mentor using **Pandas** and **SciPy** to evaluate **clinical risk factors**.
- Performed **exploratory data analysis** on specific health metrics like **resting blood pressure, cholesterol levels, and fasting blood sugar**.
- Built **Matplotlib** and **Seaborn** charts to visualize variable distributions and present findings to professors in the **UCR statistics department**.

## Projects

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### Pickleball Match Outcome Prediction | *XGBoost, K-Means, Streamlit*

[View Project](#)

- Developed a **custom web scraper** to extract match statistics, building a dataset of **over 900 matches** to **predict outcomes and identify undervalued skills**.
- Cleaned** tabular data, **created features** from shot selection, and applied **K-Means clustering** to categorize player playstyles.
- Built a **Streamlit** application to visualize win probabilities using an **XGBoost** model selected for its high performance on the data.

### Pearson Student Data Analysis | *Pandas, Scikit-learn, Matplotlib, Seaborn*

[Presentation](#)

- Awarded Best Overall Project** in the Data at UCI project competition for impactful analysis.
- Performed **EDA** with **ANOVA** and **Chi-Square** tests to identify significant factors influencing student performance, alongside clustering analysis using **K-Means** and **K-Modes**.
- Optimized a **Random Forest Regression** model using **GridSearchCV** and **cross-validation**, achieving  $R^2$  scores of **0.93** (Math Data) and **0.84** (Portuguese Data) while minimizing overfitting.

### LungLens | *Flask, TensorFlow, Google Cloud, TailwindCSS, React, Deep Learning*

<https://lunglens.vercel.app/>

- Developed a **CNN model** using **TensorFlow** and **Keras** to predict lung cancer from CT scan images, achieving **98.9% accuracy**.
- Preprocessed the **IQ-OTH/NCCD** dataset of medical images, optimizing the model to classify cases as **normal**, **benign**, or **malignant**.
- Built a frontend using **Next.js**, **React**, and **Tailwind CSS**, allowing users to upload CT images and receive real-time predictions.
- Integrated **Flask** with a REST API to handle image uploads and interact with the model.

### Fake News Detection System | *Flask, TensorFlow, Keras, NLTK*

[github.com/stutterk1d/NLPFakeNewsProject](https://github.com/stutterk1d/NLPFakeNewsProject)

- Developed a **Fake News Detection System** using an **LSTM model** in **TensorFlow**, achieving **98.75% test accuracy**.
- Built a **REST API** with **Flask** to classify news articles in real-time based on text inputs.
- Preprocessed news datasets with **NLTK** by removing stopwords, cleaning text, and tokenizing input data.
- Designed a web interface using **HTML/CSS**, allowing users to input articles and receive instant classification results.

### Spotify Song Recommender System | *Flask, scikit-learn*

[github.com/stutterk1d/SpotifyRecommender](https://github.com/stutterk1d/SpotifyRecommender)

- Developed a music recommendation system using **Flask**, **TF-IDF**, and **KMeans clustering** to suggest songs based on artist and song title similarities.
- Built a **REST API** to deliver real-time song recommendations and display results.
- Designed an interactive user interface using **HTML**, **CSS**, and **JavaScript**, providing personalized recommendations.
- Preprocessed a dataset of thousands of songs, using **TF-IDF** vectorization for feature extraction and **KMeans clustering** for grouping similar songs.

## Leadership

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### Highlander Statistics Society

**Riverside, CA**

*President*

*May 2023 – June 2024*

- Led the management and execution of two data science projects in **Python** and **SQL**, with a team of approximately five members per project.
- Supervised a "Predictive Modeling for Housing Prices" project, which involved **data preprocessing**, **exploratory data analysis**, **feature engineering**, model building, **evaluation**, and deployment of a predictive housing price model.
- Oversaw the "Customer Behavior Analysis" project, which included **data import**, **data exploration**, **customer segmentation**, **cohort analysis**, **product analysis**, **SQL** reports, and **data visualization** using **Python** libraries.

### Association of Computing Machinery at UCR

**Riverside, CA**

*Alumni Chair*

*June 2023 – March 2023*

- Organized and coordinated **networking events**, **panel discussions**, and **workshops**, connecting over 100 students with industry professionals and alumni.
- Assisted in organizing **club events**, **hackathons**, and workshops, leveraging the expertise and resources of alumni to improve the overall experience.
- Led a two-week-long **data manipulation workshop** for the ACM Data Science Series, teaching **Pandas** data wrangling, cleaning, and analysis techniques. Created an **interactive coding environment** and provided personalized feedback to create an engaging learning experience.

### Artificial Intelligence Student Collective

**Riverside, CA**

*Project Manager*

*January 2024 – June 2024*

- Led development of an **8-week workshop series** on **recommender systems**, with participants gaining hands-on experience in **collaborative**, **content-based**, and **hybrid filtering**.
- Organized **guest speaker panels** and **networking events**, fostering industry connections and innovation in **AI**.